Nilanjana Debnath

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Skills

Core ML & DL Python, PyTorch, Scikit-learn, Hugging Face Transformer.
GNNs PyTorch Geometric (PyG), Deep Graph Library (DGL), NetworkX, GraphRAG, Neo4j.
GenAI & NLP LLM orchestration, Finetuning, LoRA, QLoRA, GPTQ, PEFT, Langchain, LangGraph, PydanticAI, NLTK, SpaCy.
Data Engg PySpark, MongoDB(MQL), PyMongo, SQL, Pandas, Polars, NumPy, Amazon S3.
MLOps & Infra Databricks, Kubeflow, Jupyter, FastAPI.

Professional Experience

Data Scientist @ Aviso AI **Jun 2025 - Present**

Relationship-aware Customer Engagement Scoring

- Replaced heuristic engagement grading with a learning-based approach that predicts deal health from CRM activity and buyer-seller communication, producing consistent engagement scores across enterprise sales opportunities.
- Reformulated engagement prediction as a temporal heterogeneous graph learning problem by representing buyers, sellers, opportunities and their interactions as a connected graph, capturing relationship structure and communication dynamics unavailable in tabular data.
- Automated graph construction, feature generation, model training, threshold calibration and daily inference, generating engagement grades across **30 enterprise tenants** with data validation, monitoring, recovery, and resource optimization; supported graph workloads involving 200k+ participants and 25k+ deals.
- **Tech:** PyTorch, PyTorch Geometric, Heterogeneous GNNs, Graph Attention Networks, PySpark, Databricks, AWS S3.

LLM-based Customer Interaction Understanding

- Unified customer interaction understanding into a single LLM inference workflow, replacing independent NLP pipelines for topic extraction, meeting tagging, deal stage prediction and business entity extraction.
- Extracted discussion topics, deal stages, meeting types, competitor mentions, stakeholder roles, buyer titles and account lifecycle signals from emails and meeting transcripts using semantic embeddings and prompt-engineered LLMs.
- Automated historical and daily interaction processing to continuously enrich CRM records and generate structured Activity Intelligence across **28 enterprise tenants**.
- **Tech:** GPT, LLMs, Prompt Engineering, Semantic Embeddings, Transformers, NLP, PySpark, Databricks.

Demand Forecasting & Price Optimization

- Developed demand forecasts and pricing recommendations using historical sales and pricing data.
- Benchmarked forecasting models and selected the **Temporal Fusion Transformer (TFT)** to capture temporal patterns and business events.
- Automated feature engineering, training, and multi-horizon inference for demand forecasting for **2 tenants**.
- **Tech:** PyTorch Forecasting, PySpark, Databricks, Temporal Fusion transformer(TFT).

LLM-Powered Sales Meeting Auto-Tagging & CRM Enrichment

- Developed an LLM-powered auto-tagging pipeline to classify sales meetings into a 13-stage B2B sales-cycle taxonomy using email bodies, meeting subjects, and transcripts.
- Engineered meeting–transcript matching using host validation, temporal alignment, participant verification, and fuzzy subject matching (similarity threshold >90/100).
- Implemented confidence-gated CRM writeback using Llama-3.3-70B (Groq), with incremental processing, duplicate prevention, and no-overwrite logic to preserve manual labels and scaled across 12 tenants.
- **Tech:** LLM inference, transcript processing, fuzzy matching.

Systems Engineer @ TCS **June 2019 – July 2021**

- Analyzed product usage data using Azure Log Analytics and built KPI dashboards to identify user behavior trends and support product improvements.
- Automated Azure VM management using PowerShell and Azure Automation, reducing manual effort by **90%** across thousands of virtual machines.
- **Tech:** Azure Log Analytics, KQL, Azure Dashboards, PowerShell, Azure Automation.

Education

Institution	Degree	Duration	GPA
Indian Institute of Technology Palakkad	Master of Science By Research *(Computer Science & Engineering)*	2021 - 2025	9.25 / 10
Kalyani Govt. Engineering College	Bachelor of Technology *(Information Technology)*	2015 - 2019	8.49 / 10

Publication

Heterogeneous Continuous-Time Dynamic Graph Representation Learning (HT-Graph) code **IJCNN-2025**

- Proposed **HT-Graph**, a novel framework for scalable representation learning on heterogeneous continuous-time dynamic graphs, addressing structural and temporal dependencies for dynamic link prediction.
- Introduced a **neighbor-aware memory** and **distillation-based restarter** to preserve temporal context while removing sequential dependencies, enabling parallel training.
- Achieved **46.3% faster training** on a 2-GPU setup while improving **AP** and **AUC** over SOTA dynamic GNN baselines.
- **Skills:** Python, PyTorch, PyTorch Geometric, DGL, Dynamic GNNs, Temporal Graph Learning.

Project

LLM-Powered Temporal Graph Forecasting code

- Developed an LLM-based framework to predict future relationships in temporal knowledge graphs, addressing changing patterns and unseen entities.
- Designed a graph-to-text approach to convert graph structure and historical relationships into textual evidence for LLM-based prediction. Integrated multi-source evidence, test-time adaptation, and iterative feedback to improve prediction quality without retraining the LLM.
- **Skills:** Python, PyTorch, Large Language Models (LLMs), Temporal Knowledge Graphs.